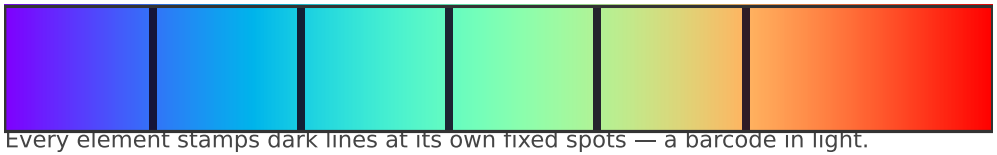
THE HUBBLE PAPER LAB — You are Hubble's night assistant (1929)

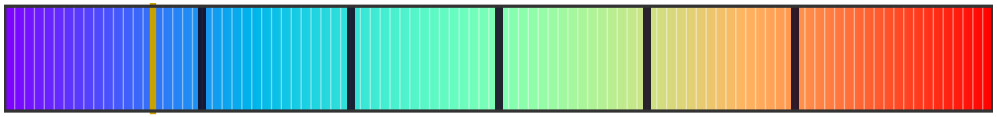
Each galaxy below shows the SAME barcode as the lab — but SHIFTED. In every panel, the GOLD TICK marks where line #1 sits at rest.
Count grid squares from the tick to the dark line above it: RIGHT = shifted toward RED = racing away · LEFT = toward BLUE = approaching. SPEED = squares × 10 km/s.

LAB FINGERPRINT — hot gas in the lab, at rest

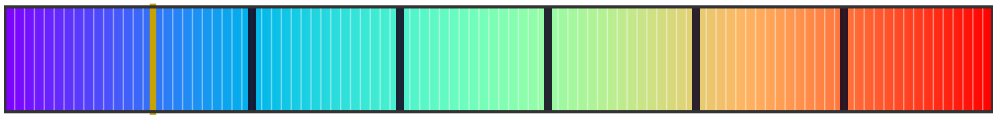


Every element stamps dark lines at its own fixed spots — a barcode in light.

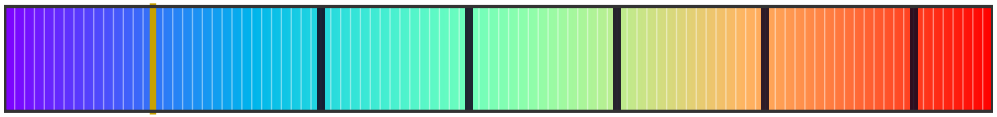
Galaxy A



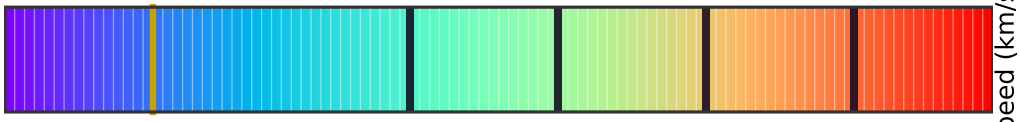
Galaxy B



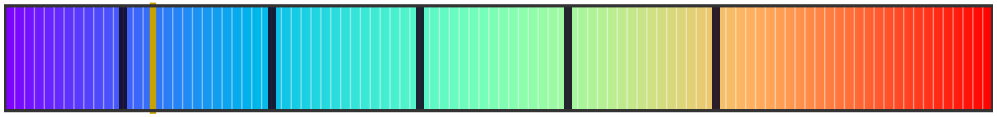
Galaxy C



Galaxy D



Galaxy E

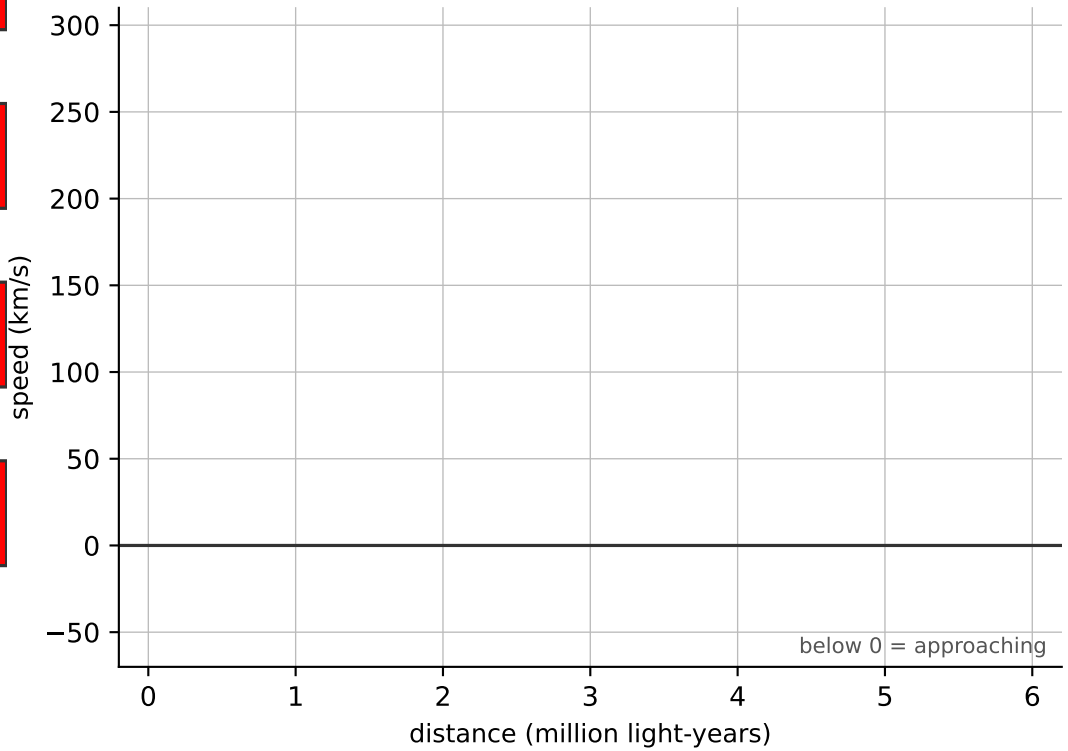


|← 1 square →|

YOUR LOG — fill every empty box

Galaxy	squares (you count)	speed (squares × 10)	distance (million ly)
A			1.0
B			2.0
C			3.5
D			5.5
E (the near one)			0.5

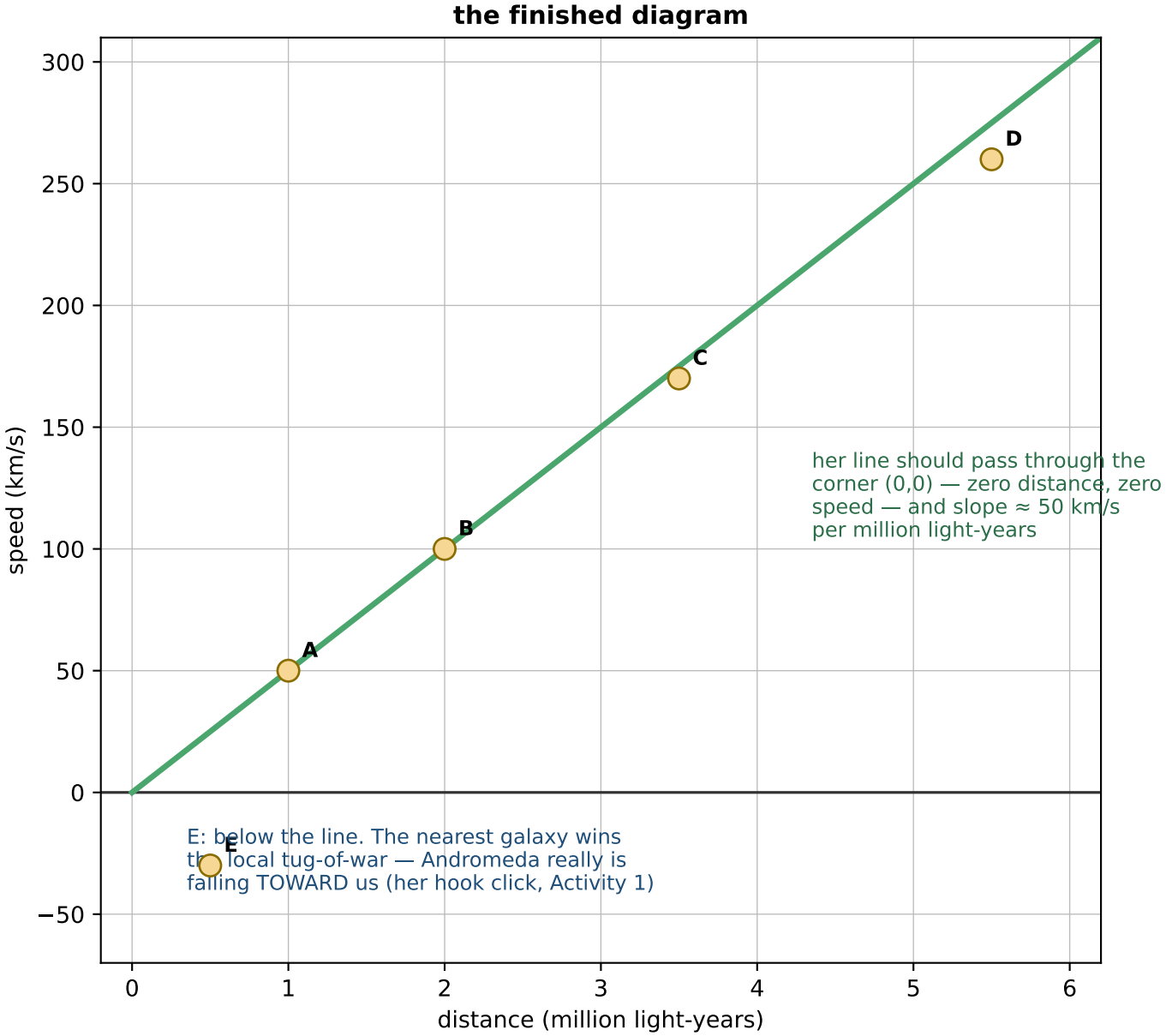
THE HUBBLE DIAGRAM — plot your 5 dots, then draw ONE line



WHEN YOUR LINE FITS: put your pencil on the far end and trace it BACKWARDS.
Where do all five galaxies end up, and how long ago? → That point is what you
(EVER-expanding? In 1998 they found the stretching is speeding up. Case still o

TEACHER KEY — the Hubble Paper Lab

Squares counted (line 1 vs gold tick): A=5 → 50 km/s · B=10 → 100 · C=17 → 170 · D=26 → 260 · E=3 to the LEFT → −30 (approaching). Distances were pre-measured with Leavitt's



TALKING POINTS

1. Every dot except E lies on ONE line through the origin — let her draw it herself, badly at first, then better. The corner matters: a galaxy at zero distance has zero speed. A straight line here is a law, not a coincidence.

2. E is the honesty dot: expansion is a large-scale average; locally gravity can win.

3. RUN THE FILM BACKWARDS: trace the line to the corner — everything converges. ~13.8 billion years. Same line, reversed.

4. Slope ≈ 50 km/s per million ly in this teaching model (the real Hubble constant is ~70 per Mpc ≈ 22 per Mly — say 'model' if asked).

5. The textbook page never does this experiment — that was Red Pen catch #2. Today she did what the title promised.

6. If she asks 'who measured the distances?': Leavitt's law. If 'who measured the shifts?': mostly Slipher, before Hubble.